

Inference labs use the five-step template: Hypothesis → Visualise → Assumptions → Conduct → Conclude.

Learning objectives

- Distinguish correlation from regression and say when each is appropriate.
- Fit Model-I (OLS) and Model-II (major-axis / SMA) lines and interpret the difference.
- Report an association with a slope, interval, and p -value rather than with a correlation alone.

Prerequisites

Hypothesis testing and basic R.

Background

Correlation and regression answer related but different questions. Correlation asks *how tightly do two variables track each other?* and is symmetric in X and Y. Regression asks *how does the mean of Y change with X?* and treats the two axes asymmetrically: X is assumed to be measured without error and Y is modelled. This asymmetry matters for prediction (which needs a regression line) and for comparing slopes across groups (which also needs regression), but it becomes problematic when both variables carry measurement error and neither can be declared the predictor.

Model-II regression — of which the standardised major axis (SMA) and reduced major axis are the common flavours — was developed for exactly this case. When both X and Y are measured with error, ordinary least squares (OLS) is biased toward zero, because it attributes all residual variation to Y. Model-II methods acknowledge error in both directions and give a less biased estimate of the true functional relationship. The price is a weaker theoretical grounding for inference and the need to report the method used.

Model-II regression has a respectable history in allometry and in method-comparison studies, where the instruments on both axes are imperfect. In a clinical lab context you will often see Deming regression for comparing two assays; SMA and Deming are close cousins under reasonable assumptions. The choice should follow the science: *is X a fixed, controlled input, or a noisy measurement of something?*

Setup

```
library(tidyverse)
library(broom)
set.seed(42)
theme_set(theme_minimal(base_size = 12))
```

1. Hypothesis

We will simulate two noisy measurements of the same underlying quantity — say, body size measured by two instruments — and ask how OLS and SMA differ.

Null hypothesis: the two measurements are uncorrelated. Working alternative: they are positively associated, with a slope near 1 if the instruments agree.

2. Visualise

```
true_size <- rnorm(n, mean = 50, sd = 10)
x <- true_size + rnorm(n, 0, 3)
y <- true_size + rnorm(n, 0, 3)
dat <- tibble(x, y)

ggplot(dat, aes(x, y)) +
```

```
geom_point(alpha = 0.6, colour = "grey30") +
labs(x = "Instrument 1", y = "Instrument 2")
```

A cloud of points that climbs from lower-left to upper-right. Both instruments carry error, so neither deserves to be called the truth.

3. Assumptions

OLS assumes X is measured without error and residuals in Y are approximately normal with constant variance. SMA assumes errors in both variables are comparable in scale. Both assume linearity.

```
par(mfrow = c(1, 2))
plot(fit_ols, which = c(1, 2))
par(mfrow = c(1, 1))
```

Residuals look patternless; the QQ plot is close to the diagonal. The OLS diagnostics are fine; they just do not capture the fact that X is also noisy.

4. Conduct

```
fit_ols <- lm(y ~ x, data = dat)
tidy(fit_ols, conf.int = TRUE)
```

Now a hand-coded SMA slope. The SMA slope is the ratio of standard deviations, signed by the correlation:

```
sma_intercept <- function(x, y) mean(y) - sma_slope(x, y) * mean(x)
```

```
b_sma <- sma_slope(dat$x, dat$y)
a_sma <- sma_intercept(dat$x, dat$y)
b_ols <- coef(fit_ols)[2]
```

```
c(ols_slope = unname(b_ols), sma_slope = b_sma)
```

```
geom_point(alpha = 0.5, colour = "grey40") +
geom_abline(intercept = coef(fit_ols)[1], slope = b_ols,
  colour = "steelblue", linewidth = 0.9) +
geom_abline(intercept = a_sma, slope = b_sma,
  colour = "firebrick", linewidth = 0.9, linetype = 2) +
labs(x = "Instrument 1", y = "Instrument 2",
  title = "OLS (blue) vs SMA (red)")
```

The SMA slope is closer to 1. OLS is attenuated because it treats X as fixed when it is not.

5. Concluding statement

In simulated data ($n = 200$), instrument 2 increased with instrument 1 (Pearson $r = \text{round}(\text{cor}(\text{dat}\$x, \text{dat}\$y), 2)$; OLS slope $\text{round}(\text{unname}(\text{b_ols}), 2)$; SMA slope $\text{round}(\text{b_sma}, 2)$). Because both instruments carry measurement error, we report the SMA slope as the functional relationship and note the OLS slope for comparison.

Common pitfalls

- Reporting a correlation when a slope is what the reader needs.
- Using OLS to compare two noisy measurements and concluding the slope differs from 1 when the attenuation is an artefact.

- Forgetting that r has no units and therefore cannot substitute for an effect size on the measurement scale.

Further reading

- Warton DI et al. (2006), *Bivariate line-fitting methods for allometry*.
- Legendre P, Legendre L. *Numerical Ecology*, ch. 10.
- Altman DG, Bland JM (1983), *Measurement in Medicine*.

Session info

See also — chapter index

- Methods in this chapter
- Find your method (Chapter 0)